

XAI 509 Course Project

Wav2Vec2 CTC Fine-Tuning with Output-Distribution Regularization

LibriSpeech 1-hour Fine-tuning · Maximum Entropy vs Label Smoothing

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June 11, 2026

Outline



- Background & Problem

- Two Regularization Methods

- Experimental Setup

- Main Results

- MaxEnt λ Sweep & Analysis

- Summary

Project Overview

| Wav2Vec2 CTC Fine-Tuning + Regularization on LibriSpeech 1h

Goal

Fine-tune **Wav2Vec2** with the **CTC loss** on a small (1-hour) LibriSpeech subset and report WER on test-clean / test-other.

Core Idea

Small fine-tuning data → **over-confidence**. Compare two output-distribution regularizers: **Maximum Entropy** and **Label Smoothing**.

Key Finding

Both regularizers improve over baseline. **Label Smoothing ($\alpha=0.10$) achieves the best WER** — clean **19.51 %** (−1.65 pp), other **28.30 %** (−1.87 pp), outperforming both baseline CTC and Maximum Entropy training.

Scope / Limitation

Only $\alpha = 0.10$ swept for label smoothing; MaxEnt has 5-point λ sweep. *Single-seed experiments.*

Background

| Wav2Vec 2.0 + CTC for ASR

Wav2Vec 2.0

- Self-supervised speech representation
- Pretrained on 53k hours of unlabeled audio
- CNN feature encoder → Transformer context
- Fine-tunable for downstream ASR

CTC Loss

- No frame-level alignment required
- Marginalizes over all valid alignments
- $WER = (S + I + D) / N$
- Lower WER = better recognition

Why this project?

Fine-tuning Wav2Vec2 on just **1 hour** of speech is challenging — the model can become **over-confident on training data**, producing sharp output distributions that fail to generalize to **test-clean / test-other**.

Method

Two Regularization Strategies

Both methods penalize sharp output distributions, but apply different weighting schemes.

Maximum Entropy

$$\text{Loss} = \text{CTC} - \lambda \cdot H(p)$$

$$H(p) = -\sum p_v \log p_v$$

Properties:

- Weights penalty by predicted probability p
- Pushes confident tokens toward uniform first
- Listed in lecture 8 "Topics You Can Try"

Label Smoothing

$$\text{Loss} = (1-\alpha) \cdot \text{CTC} + \alpha \cdot \text{LS}(p)$$

$$\text{LS}(p) = -(1/V) \sum \log p_v$$

Properties:

- Weights penalty uniformly across vocab ($1/V$)
- Equivalent to $\text{KL}(\text{uniform} \parallel p)$ up to constant
- Reported as part of lab's MoChA paper

→ Closely related (Pereyra et al. 2017) but mathematically distinct.

Experiment

Setup

Pretrained model	facebook/wav2vec2-base (95M params)
Train set	LibriLight 1-hour subset (WebDataset, 5 shards)
Eval sets	test-clean (2620 utt) / test-other (2939 utt)
Training steps	2000 (~222 epochs on 1h set)
Optimizer / LR	AdamW, lr=1e-4, warmup 500 steps
Batch	16 × grad_accum 2 = effective 32, fp16
Frozen modules	CNN feature encoder
Hardware	RTX 3090 (24 GB), ~35 min per run

Each variant (baseline / MaxEnt λ / Label Smoothing α) trained with identical settings for fair comparison.

Results

Main Comparison — Word Error Rate (%)

Method	test-clean	test-other	Δ vs Base
Baseline (CTC only)	21.16	30.17	—
Maximum Entropy ($\lambda = 0.10$)	19.93	28.80	-1.37
Label Smoothing ($\alpha = 0.10$)	19.51	28.30	-1.87

Takeaways

- Both regularizers **beat baseline by > 1 pp** on both splits.
- Label Smoothing > MaxEnt by **0.42 pp (clean) / 0.50 pp (other)**.
- Larger improvement on **test-other (harder set)** — regularization helps generalization most where it matters.

Results

Maximum Entropy — λ Sweep

λ	test-clean	test-other
0.00 (baseline)	21.16	30.17
0.01	20.60	29.67
0.03	20.79	29.65
0.05	21.21	30.46
0.07	20.93	29.80
0.10	19.93	28.80

Observations

- **Small λ (0.01-0.03)** gives modest improvement
- **$\lambda = 0.05$** actually hurts performance
- **$\lambda = 0.07$** recovers, but not best
- **$\lambda = 0.10$** achieves the lowest WER

Best operating point:

$\lambda = 0.10 \rightarrow 19.93 / 28.80$

Used as MaxEnt baseline in slide 7.

Analysis

Why Label Smoothing > MaxEnt

Same family, different weighting:

- **MaxEnt:** $H(p) = -\sum p_v \log p_v \rightarrow$ weights by p_v — already-confident tokens get pushed first
- **LS:** $LS(p) = -(1/V) \sum \log p_v \rightarrow$ uniform weight across vocab — all tokens regularized equally

Qualitative example (test-clean utterance 1)

REF: ... THEORY OF MEMORY MUST ARRIVE AT ... VIRTUE ... SINGLED OUT ...

Baseline: ... THEORY OF MEMERY MUST ARIVE AT ... VERTUE ... SINGLEDOUT ...

LS $\alpha=0.10$: ... THEORY OF MEMERY MUST ARRIVE AT ... VIRTUE ... SINGLED OUT ...

→ Spelling errors (ARIVE, VERTUE) and spacing (SINGLEDOUT) corrected by Label Smoothing.

Summary

01

Two regularizers compared

Maximum Entropy ($H(p)$) and Label Smoothing (uniform target). Both target the over-confidence problem on small fine-tune sets.

02

Label Smoothing wins

Test-clean 19.51 % / test-other 28.30 %. Outperforms baseline by -1.65 / -1.87 pp and MaxEnt by -0.42 / -0.50 pp.

03

Bigger gains where it hurts

Improvement is larger on the harder test-other split — regularization helps generalization most under acoustic mismatch.

Future: α sweep for Label Smoothing, SpecAugment data augmentation, LM shallow fusion.

Thank You

Q & A

Code & results: [/run/models/](#), [/run/results/](#)